

Aadiy Khan

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EDUCATION

Vellore Institute of Technology, Bhopal

2023 – 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.55/10.0

EXPERIENCE

Summer Intern, Ordnance Clothing Factory (OCF), Shahjahanpur

May 2026 – Present

Ministry of Defence Summer Internship Program

Shahjahanpur, India

- Selected for the Ministry of Defence Summer Internship Program to develop AI-based solutions for automated textile quality inspection.
- Built an end-to-end computer vision pipeline using Python and PyTorch for multiclass fabric defect detection and classification.
- Trained and optimized ResNet-based models on 96,000+ industrial fabric images, achieving **75% classification accuracy** across six defect categories.
- Implemented GPU-accelerated training and Grad-CAM explainability to improve model performance and interpret defect predictions.

PROJECTS

Transformer From Scratch (Attention Is All You Need) [\[GitHub\]](#) | *PyTorch, NLP, Transformers*

2026

- Implemented the complete Transformer architecture from the original “Attention Is All You Need” research paper from scratch.
- Implemented Multi-Head Self-Attention, Positional Encoding, Layer Normalization, Residual Connections, Encoder and Decoder blocks without relying on high-level transformer libraries.
- Developed training and inference pipelines for sequence-to-sequence learning tasks.
- Gained practical understanding of attention mechanisms, representation learning, and modern LLM foundations.

Real-Time Network Anomaly Detection System (NADS) [\[GitHub\]](#) | *TensorFlow, Kafka, Docker*

Feb 2026

- Developed an end-to-end MLOps pipeline for real-time network anomaly detection using an Autoencoder Neural Network and Apache Kafka.
- Curated and transformed 2.5M+ CIC-IDS2017 network flow records, performing cleaning, encoding, and normalization across 78 features.
- Designed and trained an unsupervised Autoencoder with bottleneck representation learning, enabling detection of previously unseen attack patterns through reconstruction-error based anomaly scoring.
- Engineered a streaming inference system with Kafka producers and consumers, achieving low-latency real-time anomaly detection on incoming network traffic.

FlowBERT: Multi-Task Customer Query Routing [\[GitHub\]](#) | *TensorFlow, DistilBERT, NLP* Apr 2025 – May 2025

- Fine-tuned DistilBERT for simultaneous department classification and urgency prediction using a multi-task learning framework, reducing manual ticket triage effort by 75%.
- Designed shared Transformer encoder layers with task-specific classification heads for multi-output prediction.
- Expanded the training corpus from 50 to 600+ samples through augmentation and class-balancing strategies.
- Developed an interactive Streamlit application for real-time inference and confidence score visualization.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Keras, Pandas, NumPy

Deep Learning: CNNs, RNNs, LSTMs, Autoencoders, Transformers, Transfer Learning, Attention Mechanisms

Natural Language Processing: BERT, DistilBERT, HuggingFace Transformers, Embeddings, Text Classification

Computer Vision: OpenCV, Image Processing, Object Detection

MLOps & Deployment: Apache Kafka, Docker, Streamlit, Real-Time Inference Pipelines

Tools: Git, RDKit, NLTK, Gensim

ACHIEVEMENTS

- Selected for OCF Shahjahanpur Summer Internship Program 2026 focused on AI-based defect detection for military clothing fabric.
- Reproduced the architecture proposed in the “Attention Is All You Need” paper, including Multi-Head Attention and Encoder-Decoder training pipelines.
- Solved 350+ algorithmic problems on LeetCode.
- IBM AI Engineer Professional Certificate.
- NPTEL Introduction to Machine Learning.